



RB004 Week 5 — English Worksheet

Topic: Power-Ups — Speed Pads with `task.delay` · **Course:** Arena Legends · **Time:** about 45 minutes

This week you built a **power-up**: a speed pad that makes a player faster for a few seconds, then disappears and **respawns** later. The magic words are `task.delay` (do something later, don't wait) and `task.wait` (stop and wait here).

1 · Predict 🧠

Read each script. Before you imagine running it in Roblox Studio, write what you think will happen.

```
local pad = script.Parent

pad.Touched:Connect(function(hit)
    local humanoid = hit.Parent:FindFirstChild("Humanoid")
    if humanoid then
        humanoid.WalkSpeed = 32
    end
end)
```

A player steps on the pad. What happens to their speed? Now a loose brick falls on the pad — does anything happen? Why?


```
humanoid.WalkSpeed = 32
print("Boost on!")
task.delay(5, function()
    humanoid.WalkSpeed = 16
end)
print("This line runs!")
```

In what order do the two prints appear? Does the script stop and wait 5 seconds between them? When does the speed go back to 16?


```
pad.Transparency = 1
pad.CanCollide = false
task.wait(10)
pad.Transparency = 0
pad.CanCollide = true
```

What does the player see right after this code starts? What happens after 10 seconds? Why do we change **CanCollide** too, not just **Transparency** ?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The speed boost should last **5 seconds**, then the player goes back to normal speed (16). Right now the boost is **forever**.

```
pad.Touched:Connect(function(hit)
    local humanoid = hit.Parent:FindFirstChild("Humanoid")
    if humanoid then
        humanoid.WalkSpeed = 32
    end
end)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Same goal: boost to 32, then back to 16 **after 5 seconds**. But this boost lasts **0 seconds** — the player never feels faster.

```
if humanoid then
    humanoid.WalkSpeed = 32
    humanoid.WalkSpeed = 16
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — After a player grabs the pickup, it should hide, then **come back after 10 seconds**. Right now it disappears and **never comes back**.

```
pad.Transparency = 1  
pad.CanCollide = false
```

Hint: the script needs to **wait**, then turn the pad back on.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your arena in Roblox Studio. Build a speed pad power-up: it boosts `WalkSpeed`, resets after 5 seconds, hides when touched, and respawns after 10 seconds. When you finish, come back here.

Send a photo or video of your power-up working, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

When a player touches the pad, ...

I used **task.delay** so that ...

The pad respawns when ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you playtest your power-up in Roblox Studio. Talk through it like you are teaching someone who has never seen it. Try to use these words: **Touched, Humanoid, WalkSpeed, task.delay, respawn.**

1. Show your speed pad in the arena, then press Play.
2. Step on the pad and show how fast you run. Say what number `WalkSpeed` became.
3. Wait and show the speed going back to normal. Explain why the script did not freeze while waiting.
4. Show the pad disappearing and coming back. Say in your own words what `task.delay` does.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.